Programming Questions (Coding/Logic-based Tasks)

1. Write a Java program to filter out non-null values from a list.
2. Write a multithreaded Java program that avoids deadlock.
3. Create a Customer class and return the customer name with the second-highest salary.
4. Write a program to calculate the factorial of a number.
5. Compare two HashMap<Object, Object> instances and print their common and differing key-value pairs.
6. From two maps with Employee objects, group differences and similarities based on salary.
7. Write a Java 8 program to find the department with the maximum number of employees, considering only those with salaries greater than 1000.
8. Remove duplicate elements from an array int[] a = {1,1,2,2,2,3,4,4,5,5} without using built-in collections or Set; use only for loops and if-else.
9. Write a program to find the frequency of each character in a string.
10. Given an employee list, create a copy with an increased salary without modifying the original list.
11. Given a list ["a", "b", "c"], print all possible pattern combinations.
12. Write Java 8 code to sort employees by rating and then by name.
13. Write Java 8 code to group employees by rating.
14. Write a recursive program to calculate the factorial of a number.
15. Given a list of strings List<String> words = ["abc", "def", "abc", "xyz", "def", "ghi"], return a frequency map.
16. Design an ATM class with balance inquiry, deposit, and withdrawal functionalities.
17. Write pseudocode to check for duplicates in a singly linked list.
18. Write pseudocode to detect duplicates in a cyclic linked list.
19. Create an immutable Employee class with fields id, name, and a list of experiences.
20. Write pseudocode to retrieve the nth node from the end in a singly linked list.
21. Create a POJO class for a stock exchange entity with id, price, and a list of exchanges.
22. Write a permutation logic for a character array.
23. Find the missing number in the sequence: 4, 11, 19, 41, ?, 161.
24. Write a Java 8 program to filter and sort only NSE stocks.
25. Write a program to reverse a string and an integer.
26. From a list of words from a newspaper, print the top 10 most frequent words using Java 8.
27. Serialize a Java object into a JSON object using Java code.

Theoretical and Conceptual Questions

Java Concepts

1. What is the difference between Comparable and Comparator in Java?
2. What is serialization and deserialization?
3. How do you serialize and deserialize a bearer token?
4. Explain the difference between ArrayList and HashMap. When should you use each?
5. What are the differences between TreeSet and HashSet?
6. What are the types of class loaders in Java?
7. Why is String immutable in Java?
8. What is the difference between linked list and array list?

9. When are static and instance variables initialized?
10. Explain the concept of thread safety in Java.
11. How can you make a class immutable?
12. What are the differences between abstract classes and interfaces? When to use each?
13. How does the hashCode() and equals() implementation affect collections?
14. How does a HashMap work internally?
15. What is the internal working of HashSet?
16. What is data encapsulation and abstraction?
17. What is inheritance? Give a real-world project example.
18. What is polymorphism and how is it achieved with abstract classes?
19. What are checked and unchecked exceptions?
20. What is the final keyword in Java, and when is it used?
21. What is a POJO?
22. What are the differences between @Component, @Configuration, and @Bean in Spring?
23. What is a functional interface?
24. Explain lambda expressions and their benefits.
25. What is the Optional class in Java 8?
26. What is functional programming?
27. How does the Java Virtual Machine (JVM) work internally?
28. What is the difference between stack and heap memory?
29. What is the string pool and how does it work?
30. What are the benefits and uses of SCP (String Constant Pool)?
31. What is a thread? Explain the thread lifecycle.
32. What are the differences between synchronization and concurrency?
33. What is synchronizedMap() vs. ConcurrentHashMap?
34. What is super and this keyword in Java?
35. What is big O notation?

Spring & Microservices

1. What is dependency injection and Inversion of Control (IoC)?
2. How do you implement your own IoC container?
3. Explain the Spring bean lifecycle.
4. What is a controller in Spring Boot?
5. What modules have you worked on in Spring Boot?
6. How do you serialize data between microservices?
7. What are your roles and responsibilities in Spring Boot-based projects?
8. What is the difference between monolithic and microservice architectures?
9. What are the common design patterns used in microservices?
10. Explain the Circuit Breaker design pattern.
11. What is the Saga pattern (both scenarios)?
12. How do you troubleshoot out-of-memory errors in a production environment?
13. Explain how logging is configured in your project.
14. What are some ways to handle production issues?
15. What is API Gateway?
16. What is request routing and rate limiting?
17. What is load balancing? How is it different from rate limiting?
18. How do you configure deployment with Spring Boot and Jenkins?

Security & Authentication

1. How is security implemented in your application?
2. What is authentication and authorization?
3. How does LDAP work for security?
4. What is multi-factor authentication? How do Microsoft and Google Authenticators work?
5. What is OAuth and how do you handle bearer tokens?

System Design & Architecture

1. How do distributed systems work in your project?
2. Explain the architectural pattern used in REST APIs.
3. How does request/response processing happen in REST APIs internally?
4. How is a REST method failure handled in your application?
5. How are transactions handled in sync and async systems?
6. What is your current architecture overview?
7. What is the Abstract Factory design pattern?
8. How do you handle exceptions and memory management in microservices?

DevOps, Build, and Tools

1. What are the steps in your build process?
2. What developer tools do you use?
3. How do you resolve dependency conflicts?
4. How do you work with AWS services?

Miscellaneous & Project-Based

1. Describe your recent project and technologies used.
2. What were the challenges you faced in your recent project?
3. What are your daily work processes?
4. What is your role in the project?
5. What is your experience in the healthcare and fintech domains?
6. What are the differences between Set and List? What does set.add() do?
7. What considerations are made when choosing a key in a map?
8. Given a feed with stock data from two sources (IDs, names, prices), how do you efficiently merge them?
9. What are some common production issues you've solved?